Balla Sandeep Sankar

Roll No.: 24CSB1A05
B.Tech
Computer Science Engineering (AI DS)
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Education

- **National Institute Of Technology, Warangal** *2028*
  Computer Science Engineering (Artificial Intelligence and Data Science) CGPA: 9.00/10.00
- **Sri Chaitanya College of Education** *2024*
  Board of Intermediate Education, Andhra Pradesh Percentage: 97.7
- **St. Joseph's English Medium High School** *2022*
  Board of Secondary Education, Andhra Pradesh Percentage: 97.33

Personal Projects

- **VaultMind — Enterprise AI Knowledge Base** *May 2026 – June 2026*
  RAG-Based Document Intelligence API with Role-Based Access Control and Citation Engine
  - Tools & technologies used : Python, FastAPI, Pydantic, PostgreSQL, SQLAlchemy, JWT, bcrypt, Pinecone, SMTP, Voyage AI
  - Built a production-grade REST API with JWT authentication and department-based RBAC for secure document access. Engineered a RAG pipeline using Pinecone that improved multi-turn retrieval accuracy and generated citation-backed responses.
- **FreshSense AI — Agentic Food Freshness & Recipe Recommendation** *March 2026 – April 2026*
  Computer Vision Pipeline for Real-Time Fruit Detection, Freshness Regression, and Shelf-Life Tracking
  - Tools & technologies used : Python, YOLOv8x, EfficientNet-B0, TensorFlow, OpenCV, NumPy
  - Developed a computer vision pipeline using YOLOv8x and EfficientNet-B0 for fruit detection and freshness estimation. Implemented shelf-life tracking with automated spoilage alerts and recipe recommendations to reduce food waste.
- **IQI-GWO Vision — Metaheuristic Feature Selection for Image Classification** *October 2025 – December 2025*
  Feature Selection Using Improved Quantum-Inspired Grey Wolf Optimization
  - Tools & technologies used : Python, Scikit-learn, Random Forest, Metaheuristic Optimization (Grey Wolf Optimizer), Quantum - Inspired Algorithms, Image Processing, Feature Selection
  - Designed an IQI-GWO-based feature selection framework that reduced feature dimensionality by **27–66%** on the **Smoke Detection**, **LC25000 (Lung & Colon Cancer)**, and **Type of Plants** image classification datasets. Improved classification accuracy from **72.45–96.64%** to **96.28–99.80%** across three benchmark datasets using IQI-GWO feature selection and Random Forest classification.

Technical Skills and Interests

Languages : Python, C++, R
Frameworks & Libraries : FastAPI, Pydantic, SQLAlchemy, TensorFlow, Scikit-learn, LangChain, LangGraph, OpenCV, NumPy, Pandas, Matplotlib
Databases & Vector DBs : MySQL, Oracle, Neon PostgreSQL, Pinecone
Tools & Cloud : Git, GitHub, Docker, VS Code, Jupyter Notebook, Google Colab, Render, Vercel, Voyage AI, SMTP
Soft Skills : Problem Solving, Analytical Thinking, Communication
Coursework : Data Structures & Algorithms (C++) , Machine Learning, Database Management Systems, Operating Systems, Discrete Mathematics, Probability & Statistics
Areas of Interest : Generative AI Applications, Deep Learning , Computer Vision, AI Agents & Automation, Full-Stack ML Applications

Positions of Responsibility

- **Strategic Liaison,** Independent EdTech Deployment *June 2026*
  •Acted as the liaison between an independent developer and school leadership to pitch and pilot an AI/OCR exam-digitization tool.

Achievements

- **LeetCode :** Solved **620**+ coding problems; **Peak Contest Rating: 1717**. *Ongoing*
  Secured **Global Rank 2654** among **40,636 participants** in **LeetCode Weekly Contest 496**.